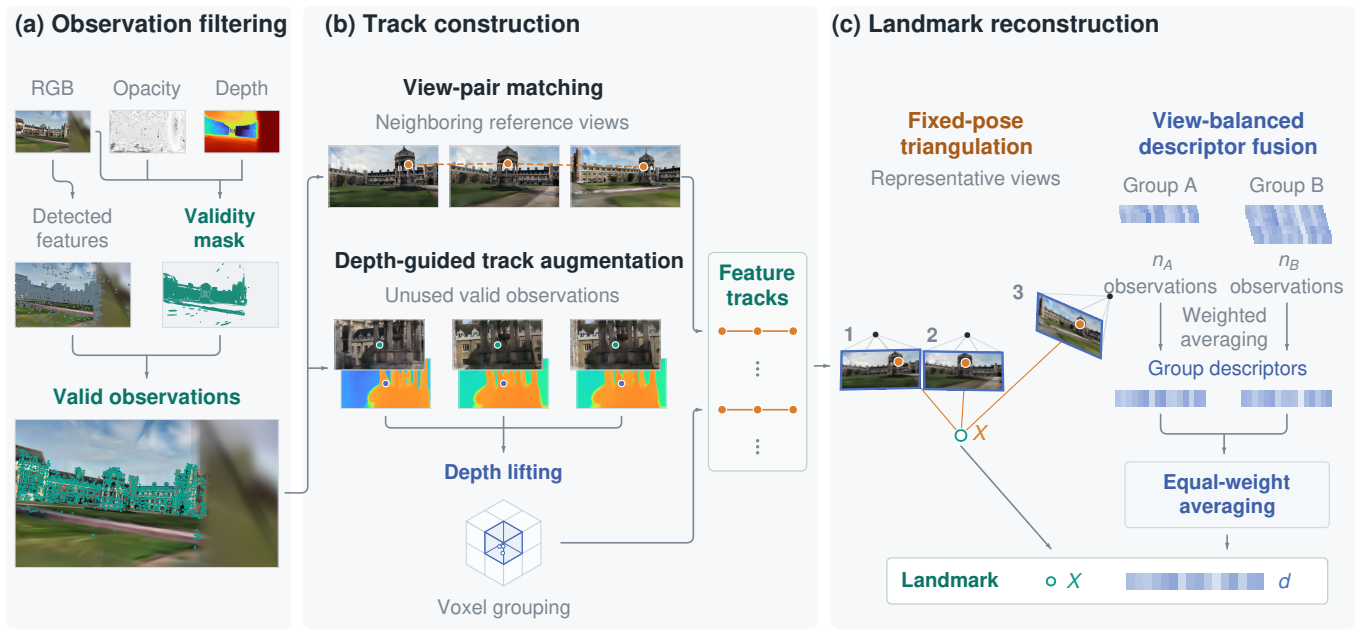

(a) Observation filtering
RGB
Opacity
Depth
Detected features
Validity mask
Valid observations
(b) Track construction
View-pair matching
Neighboring reference views
Depth-guided track augmentation
Unused valid observations
Depth lifting
Voxel grouping
Feature tracks
(c) Landmark reconstruction
Fixed-pose triangulation
Representative views
1
2
3
X
View-balanced descriptor fusion
Group A
Group B
$n_A$ observations
$n_B$ observations
Weighted averaging
Group descriptors
Equal-weight averaging
Landmark
X
d